

浙江财经大学 2019~2020 学年第一学期

《数据库原理及应用（英）》课程期末考试试卷（A 卷）

参考答案及评分标准

考核方式： 闭 卷

考试日期：2020 年 1 月 日

适用专业、班级：17 电子商务

题 号	一	二	三	四	五	六	七	八	九	十	总分
得 分											
评卷人											

(共 7 大题)

1. Questions: (Points: 15)

(1) What is a data model? Discuss three components of data model?(7)

A data model is an integrated collection of concepts for describing data and manipulation on the data, relationships between data, and constraints on the data.(4)

A data model has three components:

- a structural part: A description of the data and relationships between the data.(1)
- a manipulative part: Manipulations on the data (the operations allowed).(1)
- a set of business rules. (1)

(2) What is view? What are the advantages of the view?(8)

The dynamic result of relational operations operate on one or more than one base relations to get a new relation. A view is a virtual table that does not necessarily exist in the database but is generated by the DBMS from the underlying base tables whenever it's accessed. (4)

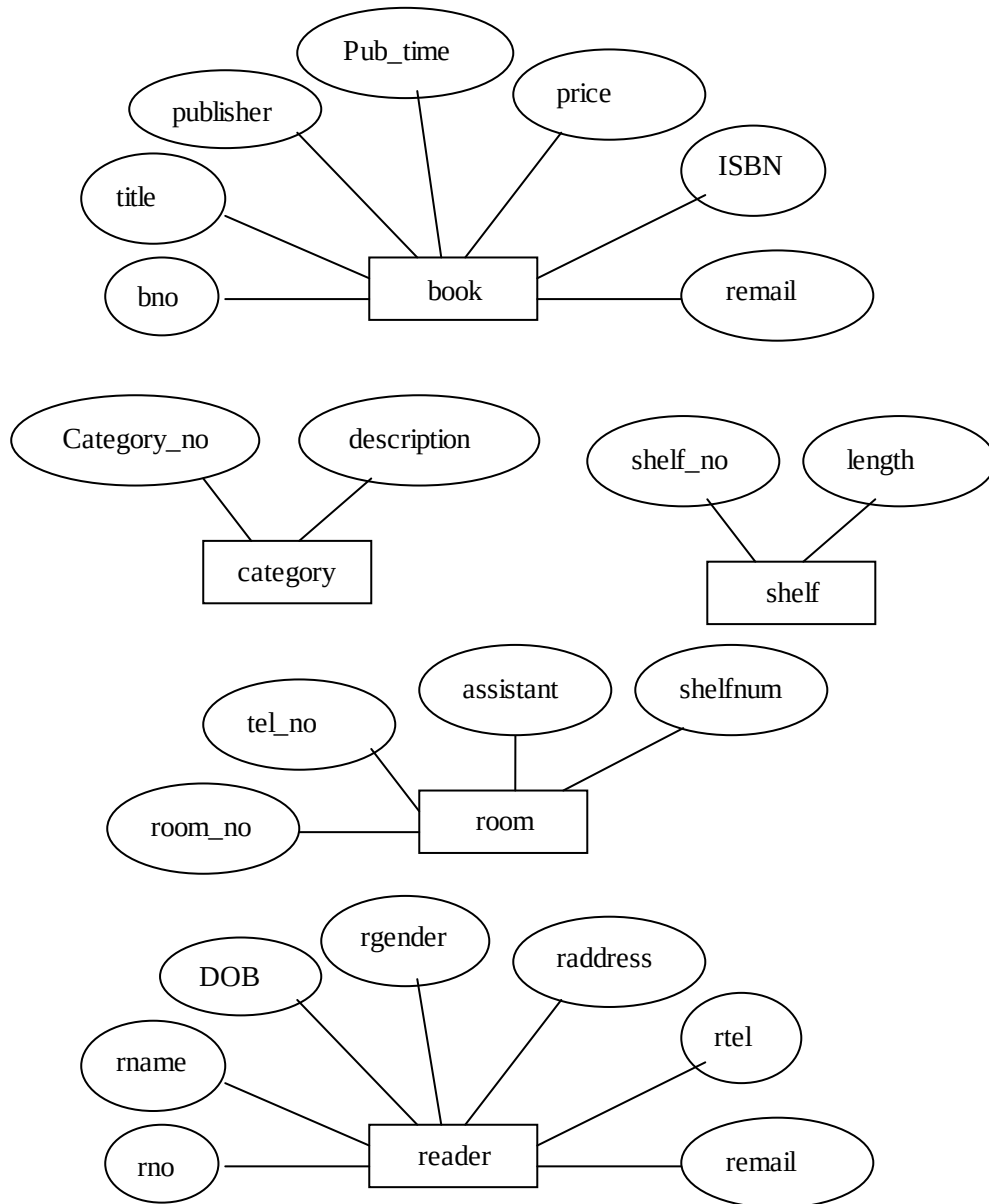
The advantages of view include:

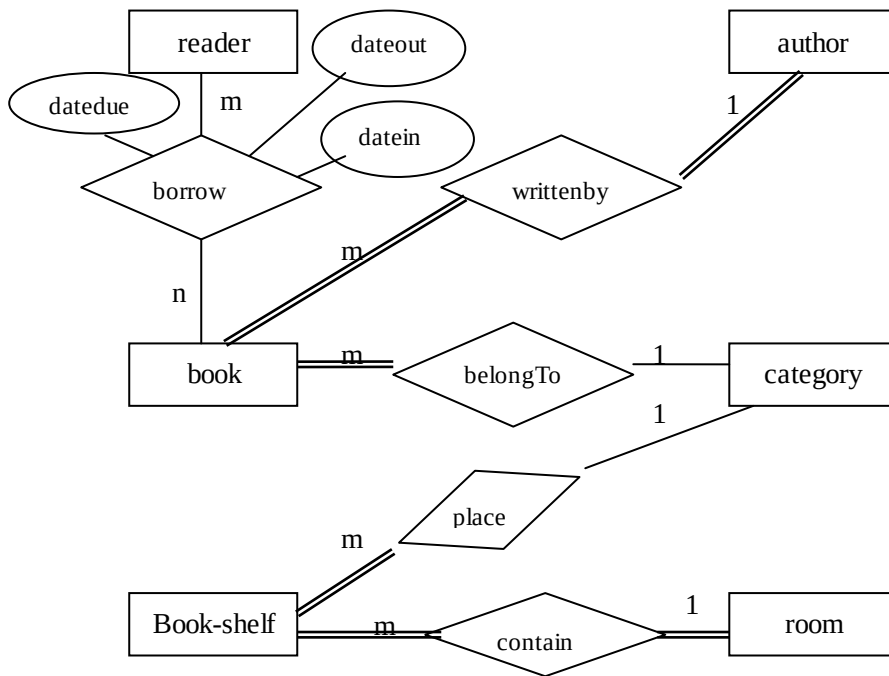
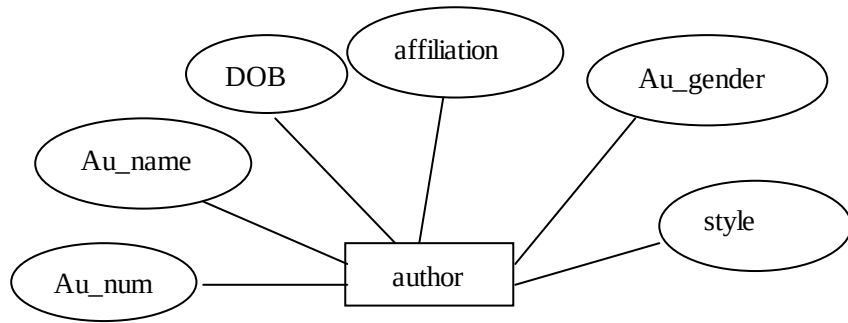
- Provide customized view of the database;(1)
- Provide logical independent;(1)
- Provide a kind of security approach;(1)
- Simplify the programming;(1)

2. For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet. (Points: 8)

(1) C (2)C (3)B (4)A(5)C (6)D (7)C (8)C

3. Given the following descriptions, create an appropriate ER diagram for the management of the Library. (point:12)





4. Calculation of relational algebra (2 points each, total 10 points).

Suppose we are given relation **R** and **S** as below, please give results of the following relational algebra expressions.

T			U		V	
A	B	C	B	D	C	E
1	2	3	1	3	3	1
1	1	3	2	4	3	2
2	1	4	1	5	4	6
2	2	4	2	6	4	7
3	4	5	1	9		
3	1	4	2	7		

1) $\Pi_B(T) \cap \Pi_B(U)$

B
1
2

2) $\Pi_C(T) \cup \Pi_C(V)$

C
3
4
5

3) $\Pi_C(T) - \Pi_C(V)$

C
5

4) $\sigma_{TC=3} (T \bowtie U)$

A	B	C	D
1	2	3	4
1	2	3	6
1	2	3	7
1	1	3	3
1	1	3	5
1	1	3	7

5) $\sigma_{TC=V,C=3}(T \times V)$

A	B	T.C	V.C	E
1	2	3	3	1
1	2	3	3	2
1	1	3	3	1
1	1	3	3	2

5. Map the following ER diagram into relation schemas.(point:10)

(1) Map the entities:

Student(Snumber,Name,Address,telephone,Birthdate,sex,Email_ID)

Primary key: Snumner;

Candidate key: Email_ID

Department(Dnumber,DName,Daddress,Dtel,Campus,office-address)

Primary key: Dnumber

Course(Cnumber,CName,Ctype,Ccredit,department)

Primary key: Cnumber

Lecture(LectureNo,Name,Major,DOB,Title,Email)

Primary key: LecutureNo

(2) Map the relationships

‘have’: post the primary key of ‘Department’ into ‘Lecture’ and act as the foreign key

Lecture(LectureNo,Name,Major,DOB,Title,Email,Dnmuber)

Primary key: LecutureNo

Foreign key: Dnumber references the primary key of ‘Department’

‘belong to’: post the primary key of ‘Department’ into ‘Student’ and act as the foreign key

Student(Snumber,Name,Address,telephone,Birthdate,sex,Email_ID,Dnumber)

Primary key: Snumner;

Candidate key: Email_ID

Foreign key: Dnumber references the primary key of ‘Department’.

‘Enrol in’: create a new relations ‘Enrol’

Enrol(Snumber, Cnumber, enrolTime, score)

Primary key: (Snumber, Cnumber)

Foreign key: Snumber references the primary key of ‘Student’

Cnumber references the primary key of ‘Course’

‘teach’: create a new relations ‘Teach’

Teach(LectureNo, Cnumber, classroom, weekday,time,hours)

Primary key: (LectureNo,Cnumber)

Foreign key: LectureNo references the primary key of ‘Lecuture’

Cnumber references the primary key of ‘Course’

6. (point: 13)Consider the following relations for sale activities:

Sale(Cno, Cname, Caddress, Ctel, Pno, Pname, price, Pcategory, Warehouse, Quantity)

Cno ,Cname, Caddress, Ctel represent the number ,name, address, telephone number of the customers, **Pno, Pname, price, Pcategory** represent the number, name, price, category(种类) of the products sale to the customers, for each category of product a certain warehouse is specified. We also record the quantity when product is sold. One customer can buy one or more than one products and one products can be purchased by one or more than one customers.

(a) Identify the primary key of this table; (point:2)

(b) Identify all the functional dependencies within the above relation; (point: 3)

(c) What normal form is the relation in? Explain your answer.(point: 2)

(d) Apply normalization until the relations are BCNF. State the reasons behind each decomposition. (point:6)

- (a) The primary key is : (Cno,Pno)
- (b) Functional Dependencies in this relation are listed following:
 (Cno,Pno) $\rightarrow$ Cname, Caddress, Ctel
 (Cno,Pno) $\rightarrow$ Pname, price, Pcategory, Warehouse
 (Cno, Pno) $\rightarrow$ Quantity
 Cno $\rightarrow$ Cnmae, Caddress, Ctel
 Pno $\rightarrow$ Pname, price, Pcategory, Warehouse
 Pcategory $\rightarrow$ Warehouse
- (c) This relation is 1NF, because each attribute in this relation are single valued and there are some non primary key attributes partially dependant on the primary key.
- (d)
 The 1st step: eliminate the partial dependencies:
 Remove the attributes which partially dependant on the primary key along with their determinant, and create two new relations.
 Customer(Cno, Cnmae, Caddress, Ctel)
 Product(Pno, Pname, price, Pcategory, Warehouse)
 Sale(Cno,Pno, Quantity)
- (3)
 In the above three relations, they are all 2NF. 'Customer' and 'Sale' are both 3NF, because there is no non-primary key attributes transitively dependant on the primary. 'Customer' and 'Sale' are BCNF, because in their FDs, each determinant is candidate key.
 As to the relation "Product", its FD shown as following:
 Pno $\rightarrow$ Pname, price, Pcategory, Warehouse
 Pcategory $\rightarrow$ Warehouse
 So, there is an attribute 'Warehouse' which is transitively dependant on the primary. Eliminate the transitive dependency by decompose the relation:
 Product(Pno, Pname, price, Pcategory)
 Warehouse(Pcategory, Warehouse)
 These two relation are both 3NF and BCNF, because each determinant is primary key.
- (3)

7. (point:32) **The following tables form part of a database held in a relational DBMS.**

Driver (Dnumber, Dname, Dsex, Dage, Daddress, Spv_Dnum)

Bus (Bnumber, Btype, capacity)

Schedule (RouteNo, Bnumber, Dnumber, date, d_time)

Each driver has a unique number represented by '**Dnumber**'. The name, sex, age, address of the driver represented by **Dname**, **Dsex**, **Dage**, **Daddress** respectively, and the **Spv_Dnum** is the number of the supervisor. Every bus has a unique number (Bnumber) and a type (Btype) and a capacity (=maximum number of passengers, 车子的座位数). Furthermore, we have a table that contains the scheduling of buses and drivers on existing routes (公交线路). All routes start at the central bus station. The attribute 'time' indicates the departure time (发车时间). One route has a

number of buses and drivers, and for each day, a bus and a driver can only be scheduled on one route.

(1) Identify the primary keys, candidate keys (if have) and foreign keys of the above relations.(point: 6)

The primary key of 'Driver' is Dnumber, and its foreign key is Spv_Dnum which references the primary key of itself.(2)

The primary key of 'Bus' is Bnumber.(1)

The primary key of 'Schedule' is (RoutNo, Bnumber, date), CK: (RoutNo, Dnumber, date), (Bnumber, Dnumber, date). There are two foreign keys in this relation. The first is Bnumber which references the primary key of Bus and the second is Dnumber which references the primary key of Driver.(3)

(2) Formulate the following queries in RA and SQL respectively: (point:16)

Q1: List the numbers, types of the buses with a capacity of more than 60.

$\Pi_{Bnumber, Btype}(\delta_{capacity > 60}(Bus))$
 Select Bnumber, Btype
 From Bus
 Where capacity > 60

Q2: Give the numbers, names of the drivers who are scheduled today (getdate()).

$\Pi_{Dnumber, Dname}(Driver \bowtie \delta_{date = getdate()}(Schedule))$
 Select Driver.Dnumber, Dname
 From Driver join Schedule on Driver.Dnumber=Schedule.Dnumber
 Where date=getdate()

Q3: List the numbers of the buses which is not scheduled today.

$\Pi_{Dnumber}(Driver) - \Pi_{Dnumber}(\delta_{date = getdate()}(Schedule))$
 Select Dnumber
 From Driver
 Where Dnumber not in
 (select Dnumber
 From Schedule
 Where date=getdate())

Q4: Give the numbers and names of the drivers who have driven on all routes.

$\Pi_{Dnumber, Dname, RoutNo}(Driver \bowtie Schedule) \div \Pi_{RoutNo}(Schedule)$
 Select Dnumber, Dname
 From Driver
 Where not exists
 (select *
 From Schedule S1
 Where not exists(
 Select * from Schedule S2
 Where S2.Dnumber=Driver.Dnumber
 And S2.RoutNo=S1.RoutNo

)
)

(3) Formulate the following questions in SQL: (point:10)

Q5: Give the total number of buses of the company.

```
Select count(Bnumber)
From Bus
```

Q6: Give the number of buses of on each route today.

```
Select routNo, count(Bnumber)
From Schedule
Where date=getdate()
Group by routNo
```

Q7: create a view which contains the buses numbers which have worked more than 25 days this month.

```
Create view workdays(Bnumber, days)
As
Select Bnumber, count(*)
From Schedule
Where month(date)=month(getdate())
Group by Bnumber
Having count(*)>=25
```

Q8: Give the names of the bus drivers who are driving on more than 3 routes.

```
Select Dnumber,Dname
From Driver D join Schedule S on D.Dnumber=S.Dnumber
Group by Dnumber, Dname
Having count(distinct routNo)>=3
```

Q9: Who has never driven a bus with a capacity of more than 80?

```
Select Dnumber, Dname
From Driver
Where Dnumber not in
( select distinct Dnumber
  From Schedule S join Bus B on S.Bnumber=B.Bnumber
  Where capacity >=80
)
```